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RESEARCH ARTICLE

Explainable Machine Learning for Radio Environment Mapping: An Intelligent System for Electric Field Strength Monitoring

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ABSTRACT The accurate characterization of signal propagation is critical for optimizing wireless network performance and supporting applications such as electromagnetic field (EMF) exposure assessment and the development of Radio Environmental Maps (REMs). This study proposes a novel, explainable machine learning system to predict electric field strength across diverse urban, semi-urban, and rural environments in Cyprus. The system is trained on a rich dataset comprising 6,543 EMF measurements collected in 2023 at mobile phone and digital TV stations, following CEPT/ECC/REC/(02)04 recommendations. The dataset includes geospatial and environmental features such as antenna distance, population density, urbanization level, and detailed built environment characteristics (e.g., volume, surface, and height). We evaluate multiple machine learning models—kNN, neural networks, decision trees, random forests, XGBoost, and LightGBM—using a two-semester split for training and assessment. Best performance was achieved with the Random Forest model, which yielded the lowest RMSE among all models. Gradient boosting models (XGBoost and LightGBM) also performed well, with RMSE values slightly higher than RF while offering flexible and scalable configurations. In contrast, k-NN and neural networks showed higher RMSE values, indicating they were less effective for this specific task. Across all models, confidence intervals were narrow, demonstrating stable and reliable predictions. Explainable AI techniques revealed that antenna distance, building volume, and population density are the most influential predictors of EMF intensity. Our approach outperforms traditional signal models by incorporating urban morphology and demographic context. As part of this system, we also create a Geographic Information System (GIS) that displays electromagnetic field strength maps derived from our explainable machine learning models. This contributes a scalable, interpretable framework for EMF exposure mapping to support regulatory monitoring, urban planning, and smart city initiatives.

INDEX TERMS Electric field strength, explainable machine learning, machine learning, radio environment map, WEB GIS.

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I. INTRODUCTION

A. GENERAL CONTEXT

Technological progress has become a defining feature of contemporary society, permeating nearly every aspect of daily life. Among the most transformative developments is the integration of the Internet of Things (IoT), which has enabled the interconnection of countless devices used in virtually all human activities—from home automation and wearable health monitors to industrial sensors and smart city infrastructure. As these technologies have become increasingly embedded in everyday routines, the demand for robust, high-speed, and ubiquitous wireless communication networks has surged dramatically [37]. This growth has been particularly pronounced in the rollout and adoption of mobile communication networks, especially with the global deployment of fifth-generation (5G) systems [3]. 5G networks promise not only faster data rates but also ultra-low latency and support for massive device connectivity, making them a cornerstone of future digital ecosystems. As a consequence, cities are now increasingly saturated with wireless infrastructure, such as small cells and base stations, designed to provide seamless coverage across dense urban environments. With this massive expansion of wireless networks comes a critical need for accurate models that can predict how electromagnetic signals propagate through various environments [45]. Signal propagation is influenced by a multitude of factors, including terrain, buildings, atmospheric conditions, and the spatial arrangement of antennas. Accurate propagation prediction is therefore essential for optimizing network performance, reducing interference, and ensuring quality of service. More broadly, it plays a central role across multiple scientific and engineering disciplines [35]. In environmental science, for instance, signal propagation models contribute to the creation of electromagnetic pollution maps, which help assess public exposure to radiofrequency (RF) fields [5]. These maps are increasingly relevant in the context of growing concerns about the potential health effects of long-term exposure to non-ionizing radiation. In the fields of wireless communications and radio frequency engineering, signal prediction supports the estimation of path loss functions and the development of Radio Environmental Maps (REMs), both of which are critical for network design, capacity planning, and smart city applications [19]. One concrete application is the planning and placement of base stations to maximize coverage and capacity while minimizing energy consumption and signal overlap. This is particularly vital in the context of 5G networks, which use higher-frequency bands that are more susceptible to signal degradation and require denser infrastructure [46]. Effective planning relies on precise radio maps that represent the spatial distribution of signal strength or quality across different environments. From a technical perspective, one of the central challenges in wireless communications and RF engineering is to develop models that can accurately describe how signals behave in real-world scenarios [31]. This often involves the estimation of the path loss

function—a mathematical representation of how signal power decreases with distance and environmental factors [30]. These estimations are foundational for generating various types of radio maps, including power spectrum maps, which are used in spectrum management, and cognitive radio applications [33]. The practical importance of these maps extends directly to the telecommunications industry. Network providers use radio maps to inform strategic decisions regarding the deployment of infrastructure, especially in densely populated or complex urban areas [9]. For example, a 5G provider might utilize radio maps to visualize the impact of urban geometry—such as high-rise buildings and narrow streets—on signal propagation, enabling more effective deployment strategies [39]. Similarly, cognitive radio networks (CRNs) represent a paradigm shift in spectrum management. These intelligent communication systems continuously monitor the RF environment and dynamically adapt their transmission parameters to avoid interference and maximize efficiency [20]. Radio maps are essential to the functioning of CRNs, providing the environmental awareness needed for intelligent decision-making. From an environmental and public health perspective, however, the goal is somewhat different. Here, the objective is not only to manage networks efficiently but also to evaluate the exposure levels of electromagnetic fields in residential and public spaces [34]. The growing density of wireless infrastructure has led to an increase in ambient RF fields in many living environments [13], prompting concerns among the public and researchers alike regarding the potential biological and health effects of long-term exposure [10]. Although non-ionizing radiation is generally considered safe at regulated exposure levels, scientific inquiry into its possible risks remains active and necessary. In recent years, numerous studies have sought to assess the levels of electric field strength across different environments—indoor, outdoor, rural, and urban—to establish baseline exposure levels and inform public policy [6]. Despite this growing body of research, most studies to date have focused primarily on measuring electromagnetic field strength, often using specialized equipment at fixed locations [17], [29]. While such measurements are invaluable for empirical analysis, they are inherently limited in spatial coverage and scalability. What has been largely missing from the field is the development of predictive methodologies capable of estimating field strength in unmeasured areas—particularly at a national or city-wide scale. Moreover, many existing approaches rely on traditional geospatial interpolation techniques, which may not fully capture the complexity of the variables influencing signal propagation, such as urban morphology or human population density. In this context, there is a clear need for the integration of more advanced modeling techniques, such as machine learning, which can leverage large, heterogeneous datasets and uncover non-linear relationships between environmental variables and signal strength. The adoption of explainable AI (XAI) methods further enhances this approach, providing transparency and interpretability to model predictions—an

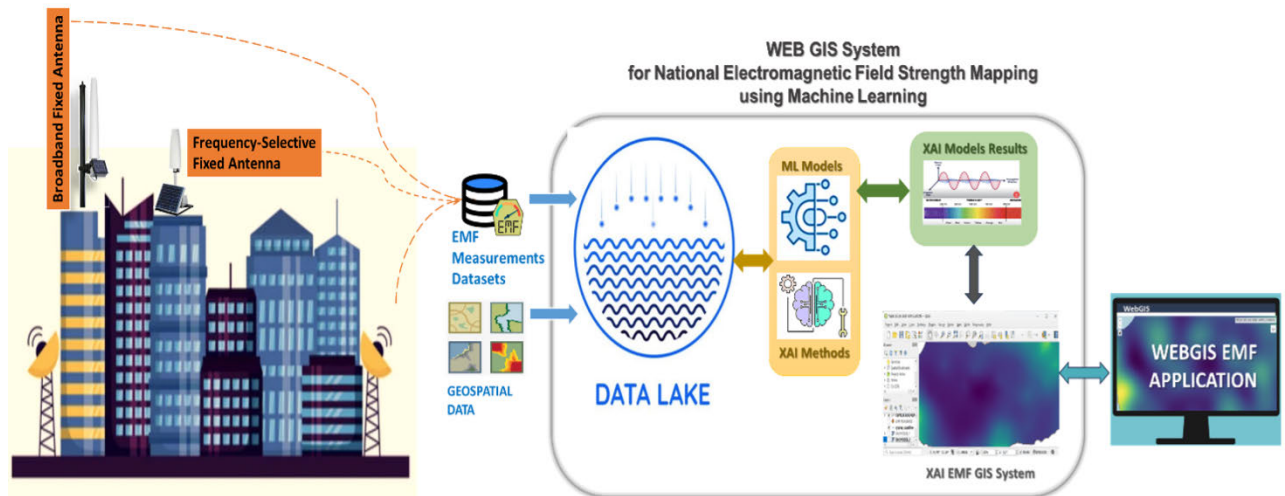


FIGURE 1. The framework of the system.

essential feature when results are to be used in public-facing platforms or regulatory decision-making.

B. CONTRIBUTION AND SCOPE

The primary objective of this study is to develop a comprehensive information system that enables the accurate prediction, interpretation, and visualization of outdoor electric field strength levels across an entire country using explainable machine learning (XAI) techniques. To the best of our knowledge, this is the first large-scale attempt to combine machine learning models with explainable AI to assess non-ionizing electromagnetic radiation exposure, and to present the results through an integrated Geographic Information System (GIS). The novelty and key contributions of our approach are threefold:

- 1) **Innovative Dataset Integration:** We introduce a unique, enriched dataset combining over 6,500 EMF measurements with geospatial and environmental variables, including population density, urbanization degree, and detailed building morphology. This multidimensional dataset enables more accurate and context-aware predictions compared to traditional signal propagation models.
- 2) **Application of Explainable AI in EMF Mapping:** Our system applies a suite of machine learning algorithms—k-NN, neural networks, decision trees, random forests, XGBoost, and LightGBM—to predict electric field strength. Through explainable AI techniques (e.g., SHAP values), we not only achieve high predictive accuracy but also identify the most influential features (e.g., antenna distance, building volume, and population density), providing actionable insights into the factors influencing EMF exposure.
- 3) **Development of a Scalable, Interpretable GIS Platform:** We create a dynamic Geographic Information System that visualizes electric field strength maps based on interpretable machine learning

outputs. This tool offers transparency, supports informed decision-making for urban planning and public health, and represents a novel integration of XAI in environmental exposure mapping.

In general, this work advances the state of the art by moving beyond black-box prediction models and offering a transparent, data-driven solution for EMF exposure analysis at a national scale.

II. RELATED WORK

A. RELATED WORK IN WIRELESS COMMUNICATIONS

In [18], the authors propose and evaluate a data-driven, model-free path loss prediction method based on a novel supervised learning approach. Exploring urban propagation, [38] introduces new input features derived from image processing tools and presents a model using a set of seven machine learning models, showing improved prediction performance. To model path loss in urban 5G environments, [15] implements a deep learning approach, combining a log-distance path loss model for line-of-sight cases with a deep learning model for nonline-of-sight scenarios. For transparency, eight characteristics were selected to model path loss using linear regression, providing insight into the behavior of the deep learning model. Further advancing path loss prediction, [30] applies a machine learning approach utilizing data from online sources such as OpenStreetMap to facilitate estimation of cell coverage in urban areas.

The study in [14] presents a path loss model that uses multidimensional Gaussian process regression (GPR) to improve spatial consistency in channel predictions, outperforming traditional models, especially with increased spatial correlation. In [12], the authors address the challenge of estimating power spectrum maps (PSMs) due to the difficulty of obtaining accurate radio propagation characteristics. To solve this, they proposed a novel PSM estimation algorithm using generative adversarial networks (GANs). Their MEGANs algorithm learns precise radio propagation features during

training rather than relying on potentially inaccurate or biased assumptions typical in traditional methods. The simulation results show that MEGANs provide significantly more accurate estimates than conventional approaches. Finally, [11] explores a deep learning-based spectral decision model for cognitive radio networks, using feature extraction to classify three levels of spectral traffic within the spectral occupancy matrix of a primary user, enhancing the efficiency of CRN. In the most recent comparable study [16], researchers evaluated 566 machine learning models in eight French cities. Six fundamental algorithms were evaluated: k Nearest Neighbors, XGBoost, Random Forest, Neural Networks, Decision Trees, and Linear Regression. The study highlighted the superior performance of the ensemble methods, particularly Random Forests and XGBoost, which consistently outperformed simpler models. Furthermore, SHAP analysis revealed substantial differences in the rank of importance of features between tree-based methods and other approaches such as k-NN, neural networks, and linear regression, highlighting the unique insights provided by the ensemble models.

B. RELATED WORK IN ENVIRONMENT

Environmental models for decision making are used in various environmental fields, and in each of these fields there are many different types of model, each incorporating various characteristics to describe the behavior of the natural system. Spatial environmental models, such as creating a map of electric strength from multiple sources, are typically the result of the interaction of multiple components with errors that do not exhibit predictable properties, making traditional hypothesis testing associated with statistical inference less suitable due to the strong assumptions often required and the difficulty, and sometimes impracticality, of testing hypotheses separately [5].

Each attempt to model the level of electromagnetic radiation has unique goals and challenges, and due to its enormous complexity, there is no ideal or standardized evaluation technique that can be applied to all models. However, machine learning, especially in recent years, has demonstrated its capabilities in fields such as physics. More specifically, in recent years, efforts have been made to utilize machine learning techniques in this context.

In [43], the research focuses on evaluating the electromagnetic field (EMF) exposure emanating from base stations (BSs) within urban city environments. The authors employ artificial neural networks (ANNs) to reconstruct EMF exposure levels based on data collected from sensors. Their approach takes into account various factors, including the spatial locations of actual BSs in the 14th district of Paris, temporal variations, and antenna orientations. A novel approach was introduced in [4], for determining the power density value, which serves as the standard for measuring human exposure to radio frequency (RF) electromagnetic fields (EMF) emitted by mmWave mobile

devices. They accomplish this using a deep learning network. Authors in [27] present an algorithm that utilizes the U-net architecture, based on convolutional neural networks, to create electromagnetic field exposure maps. This model is designed to understand the propagation patterns of wireless signals within real indoor environments, considering the various positions of Wi-Fi access points. The results of their study show that it is feasible to learn indoor propagation characteristics and environmental models from the data, ultimately leading to the generation of precise power maps to measure electromagnetic fields.

The use of a conditional generative adversarial network to accurately reconstruct the electromagnetic field exposure map based on the topology of an urban environment in the outdoors has been proposed in [28]. The authors achieve this by training the model to understand and estimate how the electromagnetic field propagates in relation to the given environment's topology. The work also compares this approach with a simpler technique called kriging where the results demonstrate that the proposed method provides precise estimates and holds promise as a solution for reconstructing exposure maps. In [42], the authors use an artificial neural network (ANN) model to perform a spatial reconstruction of radiofrequency (RF) electromagnetic field (EMF) exposure in an urban outdoor setting. They construct ANN models using input data that include details about distances to N neighboring base stations (BS), receiver locations, and temporal variations.

The proposed model demonstrates the ability to predict exposure levels effectively without becoming overly complex. An innovative artificial neural network (ANN) model presented in [40] designed to estimate Uplink transmit power is. They achieve this by using readily available parameters, including downlink connection indicators and information about the indoor environment. Using these easily accessible input features, the proposed ANN model can accurately estimate the transmit power of Uplink, achieving a mean absolute error (MAE) of 1.487 dB.

In [44], the authors conducted a study on the time and space mapping of electromagnetic field (EMF) exposure caused by cellular base station antennas (BSA). To this end, they used artificial neural networks (ANN) for this purpose. In [8] the authors presented a novel machine learning-based approach for forecasting the resulting uplink transmission power used for data transmissions based on the available passive network quality indicators and application-level information. Also, the authors of an article [41] presented a novel machine learning method based on Neural Networks (NN) to generate an electromagnetic map of radio-frequency (RF) exposure generated by WiFi sources in indoor scenarios.

III. TOOLS AND METHODS

A. DATASET

The initial data set comprises open data sourced from data.europa.eu, the official portal for European data [2], the

TABLE 1. Frequency bands and descriptions.

Code	Zone Description	Frequency
M	Radio Broadcasting Zone	87.5 - 108 MHz
VHF	Very High Frequency (excluding FM)	30 – 87.5 MHz, 108 - 300 MHz
UHF TV	Terrestrial Digital Television Zone	470 - 790 MHz
700	Mobile Telephony Zone	694 - 790 MHz
800	Mobile Telephony Zone	791 - 821 MHz
900	Mobile Telephony Zone	925 - 959.8 MHz
1800	Mobile Telephony Zone	1805.2 – 1880 MHz
2100	Mobile Telephony Zone	2110 – 2170 MHz
2600	Mobile Telephony Zone	2620 – 2690 MHz
3600	Mobile Telephony Zone	3400 – 3800 MHz
Other Frequencies	All frequency gaps within the spectrum of 30 MHz – 6 GHz not included in the above frequency zones	-

data is also available on the Cyprus National Open Data Portal, a platform that offers access to various open data sets related to the country [1]. The measurements of electromagnetic fields were performed via portable IoT devices at all installed mobile phone stations (Cyta and Epic) and at all digital television stations. The methodology used by the Department of Electronic Communications of the Ministry of Research, Innovation and Digital Policy of Cyprus is the following: Three measurements were taken at points in the environment surrounding each telecommunications station where maximum values of the exposure value to electromagnetic radiation are expected. Such points are those located at elevated positions near the telecommunications station and in the direction of the central emission axis, and the three measurements were taken at different distances each time. Additionally, the methodology used is described in Recommendation CEPT/ECC/REC/(02)04 titled “Measuring Non-Ionizing Radiation (9kHz - 300 GHz)”. Measurements of electric field intensity are made on the three propagation axes of the signal (x, y, z). For the assessment of the worst-case scenario, the electric field intensity values considered are the maximum values recorded for each emission, not the average for the 6-minute period, as provided for in the above recommendation. At each point, measurements are taken from all contributing sources of electromagnetic radiation.

The initial data set comprises 6543 measurements conducted throughout the territory of the Republic of Cyprus in 2023, furthermore, Figure 2 shows the distribution of measurement points. The frequency spectrum of the investigated radiocommunication services is shown in Table 1. The total measurements were taken in both semesters of 2023. In the first semester of 2023, 3645 measurements were taken, which will form the entire data set for model selection. The measurements of the second semester of 2023, which number 2898, will be used for model assessment of the selected models. Cyprus is the third largest island in the Mediterranean Sea and spans dimensions of 240 kilometers (149 miles) in length from one end to the other and of 100 kilometers (62 miles) width at its widest point. It is situated between latitude 34° - 36° N and longitude 32° - 35° E (Figure 2). Data processing tasks were carried out using a combination of Python libraries, which provided enhanced flexibility and a range of tools to present the results obtained.

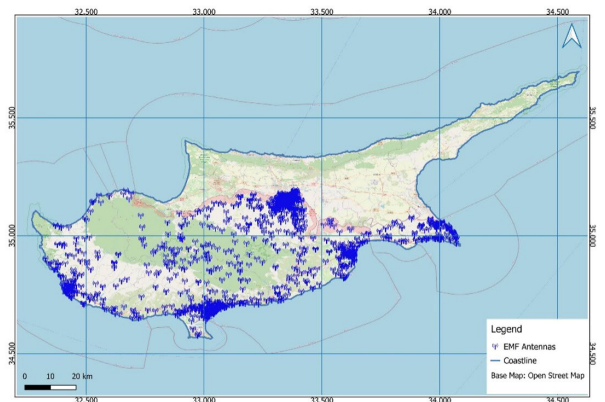
**FIGURE 2.** Dataset distribution.

Figure 3 presents a comparative box plot analysis of electric field strength (measured in V/m) across different radiocommunication services for the first and second semesters of 2023, labeled as 2023a and 2023b, respectively. Each subfigure depicts the distribution of electric field strength values for various frequency bands, ranging from low-frequency services such as FM, VHF, and UHF-TV, to higher-frequency cellular bands including 700 MHz, 800 MHz, 900 MHz, 1800 MHz, 2100 MHz, 2600 MHz, and 3600 MHz. A final category, labeled “Others,” aggregates remaining frequencies outside the primary bands. Overlaid density plots and individual data points provide additional insight into the distributional shape and variability within each band. In both semesters, cellular communication bands—particularly from 800 MHz to 2600 MHz—consistently exhibit higher electric field strength values compared to the lower-frequency bands. These cellular bands not only demonstrate elevated median values but also larger interquartile ranges and more frequent outliers, with maximum values occasionally exceeding 25 V/m. Among them, the 800 MHz and 1800 MHz bands show particularly broad distributions, which is likely a reflection of widespread LTE infrastructure deployment in both urban and semi-urban environments. This variation indicates a high level of spatial and situational diversity in signal strength for these frequencies. The 3600 MHz band, which is commonly associated with 5G networks, shows relatively lower field